



RECOMMENDER SYSTEMS

Movie Popularity and Target Audience

**Prediction Using Content-Based
Recommender Systems**

&

**Prediction of Movie Quality via Adaptive
Voting Classifier**

Gülnihal VURAT 210201001

Sema Nur AKTAŞ 210201026

Sıla EKİNCİOĞLU 220201804

Report 2 — Summary of Differences Between the Midterm and Final Work

Objective

In both the midterm and the final, the goal is the same three-part movie-analytics system: (1) a content-based recommendation engine that finds the most similar movies to a given title, (2) a model that predicts how successful a movie is from its data, and (3) a module that estimates how well a movie matches different audience age groups (Junior, Teenage, Mid-age, Senior). The midterm builds the first working version of this pipeline; the final keeps the same objective but redesigns the prediction approach for markedly better and more reliable results.

1. From multi-class to binary classification

In the midterm I split movies into six popularity classes (F, A, AA, H, SH, SDH). This fine-grained labeling was hard to learn and the results were weak. In the final I redefined the task as a binary problem — High-Quality (≥ 6.5) vs Low-Quality (< 6.5). This single change produced substantially better and more stable results and became the foundation for the rest of the redesign.

2. A new prediction method

In the midterm, my predictor was a 1D Convolutional Neural Network. I treated each movie's engineered feature vector as a one-dimensional signal and passed it through stacked Conv1D layers with batch normalization, pooling, and dropout, finishing in a softmax over the popularity classes; I also added L2 regularization, label smoothing, and early stopping to fight overfitting. Despite all this tuning, the CNN could not reach the accuracy I was aiming for: a deep network needs far more data than I had on this small, tabular dataset, and the six-way target made the problem even harder.

For the final, I designed my own Genetic-Algorithm Voting (GA-V) ensemble to take the CNN's place. The idea is to combine several diverse classifiers and let a genetic algorithm learn how much to trust each one, instead of averaging them blindly. It works in two stages:

- **Stage 1 — base learners.** I train five complementary models (Random Forest, AdaBoost, Logistic Regression, Naïve Bayes, and SVM) and collect their out-of-fold class probabilities through stratified cross-validation, so the weighting in the next stage is fit on honest, unseen-fold predictions rather than on data the models have already memorized.
- **Stage 2 — genetic optimization.** Instead of giving every model an equal vote, I assign a separate weight to each (model, class) pair and let a genetic algorithm evolve those weights. A population of candidate weight sets is scored by a fitness function (the performance of the resulting weighted vote); the best performers are selected to seed the next generation, and this repeats over many generations until the weighting converges. The trained model then predicts using these GA-found weights.

This adaptive, class-aware weighted-voting scheme is the main method I developed: it keeps the data-efficiency of classical models while letting the genetic algorithm find the best combination of them, and it clearly outperformed the midterm's CNN.

3. Simpler, more robust inputs

The midterm fed the predictor a large, heavily engineered feature vector derived from the recommendation step. In the final I moved to a compact set of direct numeric features of the movie itself, making the model easier to train and less prone to overfitting.

4. Stronger validation

The midterm relied on a single train-test split. In the final I added cross-validation — both overall and per-genre — so the reported accuracy reflects genuine generalization rather than one lucky split.

5. Streamlined recommendation engine

I kept the recommendation system but simplified it: instead of the midterm's multi-feature, multi-step similarity computation, the final uses a leaner text-similarity approach that produces comparable recommendations with far less complexity, letting the prediction model take center stage.

6. Redesigned audience-matching

In the midterm the age-group matching relied on a separate clustering-based scoring scheme. In the final I tied audience scoring directly to the GA-V model's quality prediction, blending it with genre-based demographic preferences, so the audience module is now consistent with — and driven by — my main model.

Conclusion

The defining changes are mine by design: a binary High/Low-Quality target instead of six classes, my Genetic-Algorithm-weighted voting ensemble in place of the under-performing CNN, simpler input features, and added cross-validation — together yielding clearly better and more trustworthy results than the midterm.